

Responsible AI Access to Research Repositories

An MCP Server for African Studies Data

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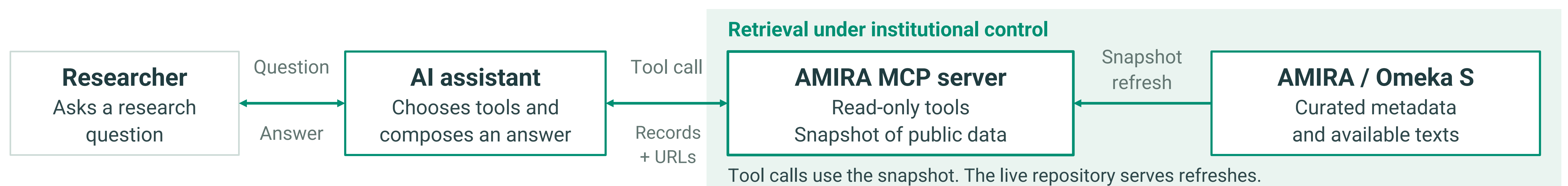
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Catalogues and digital collections at risk

Automated harvesting overloads the servers that keep open collections accessible. **Users increasingly ask chatbots rather than catalogues**, often stopping at the summary.

African collections face extraction without credit while remaining poorly represented in models. Closure and unrestricted harvesting both weaken institutions' authority over how their holdings are encountered and interpreted.

How AMIRA mediates each tool call



Africa Multiple Interactive Research Atlas (AMIRA): the Africa Multiple Cluster's research-data platform.

Around 4,000 research items across 93 projects, with source URLs. Original research data stays in partner repositories by default.

27 tools for research access

Explore the catalogue

Scope the collection, inspect facets, follow people, projects and organisations.

`get_collection_overview`, `list_subjects`, `search_persons`

Search, retrieve and cite

Research items, publications, podcasts and videos. Available full texts and transcripts, with citation exports.

`search_research_items`, `get_publication`, `search_videos`

Graphs and visualisations

MCP Apps deliver overviews, timelines and relational graphs.

Example: "What co-occurs with Islam?" using FindRelated.



Top two entities per relation type. Spoke length = shared-item count. Interactive in compatible assistants, including Claude.

MCP server as mediator

Mediated openness: a third position between closure and unrestricted extraction.

The Model Context Protocol (MCP) lets institutions define tools and the records they return. Source links preserve provenance. Tool descriptions carry curatorial expertise into the interface.

FAIR supports access. CARE asks who controls data and who benefits.

Evaluation and results

Four questions, five fresh Claude sessions each.

Topics: Islam, gender, Kenya and a misspelled name.

Query	Items	Kendall's τ	Title distance ¹
Islam	25	1.00 ± 0.00	0.040 ± 0.197
Gender	13	1.00 ± 0.00	1.080 ± 0.835
Kenya	102	1.00 ± 0.00	0.088 ± 0.284
Misspelled name	5	1.00 ± 0.00	0 ± 0

¹ Levenshtein distance between returned and source titles.

Values for τ and title distance are means ± standard deviations.

Identical result sets and ordering

All pairwise Jaccard scores: 1.00

"Rudigr Seeman" resolved to Rüdiger Seesemann in every session, returning the same five publications.

Only 24 punctuation edits and one minor title correction.

No fabricated records in these 20 test sessions.

These 20 sessions test stability and title fidelity. Relevance, interpretation and newer full-text capabilities require further evaluation.

Outlook: a locally hosted model

Next: an open-weight model hosted on Cluster or University servers as AMIRA's main chatbot interface, connected to the same MCP tools.

Server v1.17.0

27 core tools, 29 over HTTP

[Explore AMIRA](#)

[Code and install](#)



Grounded answers with AMIRA URLs for verification. MCP cannot resolve the political economy of AI extraction.

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